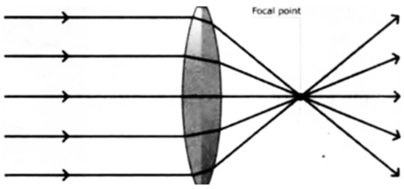
LIGHT-03

Class 08 - Science

1. In regular reflection, the angle of incidence is always equal to: **[1]**

   a) Angle of deviation
   b) Angle of refraction
   c) Angle of reflection
   d) Angle of emergence

2. The given figure is showing: **[1]**



   a) Converging of light rays.
   b) All of these
   c) Dispersion of light.
   d) Diverging of light rays.

3. Small patches of sunlight under the tree are the **[1]**

   a) Shadow of tree
   b) Images of leaves
   c) Images of Sun
   d) Shadow of leaves

4. The black board seems black because **[1]**

   a) it reflects black colour
   b) it absorbs black colour
   c) it does not reflect any colour
   d) it reflects every colour

5. A ray of light is incident on a plane mirror and angle of incidence is 80^{o}. Calculate the angle between incident ray and the reflected ray. **[1]**

   a) 80^{o}
   b) 40^{o}
   c) 160^{o}
   d) 100^{o}

6. When you stands in front of a small mirror. The image formed will appear to be of **[1]**

   a) Larger and inverted
   b) Same size and erect
   c) Smaller and erect
   d) Larger and erect

7. Due to which the air bubble seems to be sparkling in the water? **[1]**

   a) Due to reflection
   b) Due to total reflection
   c) Due to refraction
   d) Due to diffraction

8. A dentist uses ________ to see a magnified image of a teeth. **[1]**

a) Convex mirror
b) Plane mirror
c) Concave mirror
d) Concave lens

9. Danger signals are made red coloured because red light **[1]**

a) Scatter least
b) Scatter most
c) Do not scatter at all
d) Viewed in absence of light

10. Phenomenon of advance sunrise and delayed sunset is caused by: **[1]**

a) Refraction of light
b) Dispersion of light
c) All of these
d) Reflection of light

11. Which of the following phenomenon is based on the persistence of vision principle? **[1]**

a) Hemophilia
b) Myopia
c) Colour blindness
d) Cinematography

12. A narrow beam of light is made up of **[1]**

a) Only one rays of light
b) Rays of light
c) Millions of rays of light
d) Two rays of light

13. Dentists make use of ________ mirror to examine the tooth defects. **[1]**

a) convex
b) spherical
c) plane
d) concave

14. Concave lens is also known as ________. **[1]**

a) diverging lens
b) biconcave lens
c) bifocal lens
d) converging lens

15. Which of the following is an example of virtual image? **[1]**

a) Image formed on cinema screen
b) Image formed by plane mirror
c) Image formed on retina of eyes
d) Image formed pinhole camera

16. Choose the correct statement for making the images by plain mirror: **[1]**

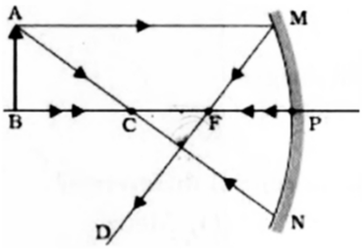
i. Image Distance = Object Distance
ii. Image Size = Object Size
iii. Laterally Converted = Erected Image
iv. Real Image

a) (i), (ii) and (iii)
b) (ii), (iii) and (iv)
c) (i), (ii) and (iv)
d) (i), (iii) and (iv)

17. Bulging outward surface of a spoon is: **[1]**

a) Convex
b) Concavo-convex
c) Plane mirror
d) Concave

18. Look at the following figure and find the position of the image **[1]**



a) Between focus and centre of curvature
b) Beyond centre of curvature
c) At pole
d) At focus

19. Which type of mirror is used as shop security image? **[1]**

a) Plane mirror
b) Convex mirror
c) Concave mirror
d) Converging mirror

20. A dark muscular structure of human eye which gives it its distinctive colour is **[1]**

a) cornea
b) eye lens
c) pupil
d) iris

21. Which metal is the best reflector of light? **[1]**

a) Copper
b) Zinc
c) Magnesium
d) Silver

22. Distance from focus to the centre of lens is: **[1]**

a) Focal length
b) Focal point
c) Radius of Curvature
d) Centre of Curvature

23. The angle of incidence is equal to the angle of incidence: **[1]**

a) Always
b) Under special conditions
c) Never
d) Sometimes

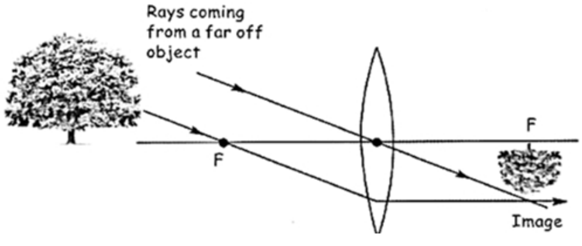
24. The angle between mirror and incident ray is 50° than the angle of reflection will be: **[1]**

a) 90^o
b) 40^o
c) 50^o
d) 45^o

25. The human eye has a converging lens system that produces an image just like a camera. If the eye views a distant object, which type of image is produced? **[1]**

a) Virtual, erect, diminished
b) Real, erect, same, size
c) Real, inverted, diminished
d) Virtual, inverted, magnified

26. When the object is far away from the convex lens, the image is ________. **[1]**



a) erect, virtual, enlarged
b) inverted, real, diminished

c) only inverted

d) inverted, real, smaller

27. A concave lens always forms an image that is ________. **[1]**

a) real, inverted and diminished

b) virtual, erect and diminished

c) real, inverted and of same size

d) real, erect and enlarged

28. Videography and cinematography is based on principle of **[1]**

a) Far point

b) Least distance of distinct vision

c) Hypermetropia

d) Persistence of vision

29. Light consists of electromagnetic waves that **[1]**

a) Can propagate in air only

b) Can propagate in rarer medium only

c) Require material medium for propagation

d) Do not require material medium for propagation

30. Images that can be obtained on screen are: **[1]**

a) Real images

b) Virtual images

c) Shadows

d) Reflected images

31. Scientist who showed white light is made up of seven colours was: **[1]**

a) Isaac Newton

b) Nicolaus Copernicus

c) James watt

d) Galileo Galilei

32. The human eye forms the image of an objects at its **[1]**

a) pupil

b) iris

c) retina

d) cornea

33. When light travels from hot air to cold air, then **[1]**

a) bends towards the normal and scatter

b) bends away from the normal

c) bends away from the normal and scatter

d) bends towards the normal

34. Bouncing back of ray of light from shining surface on which it was incident is known as **[1]**

a) refraction

b) dispersion

c) reflection

d) bending

35. A ray of light can reflected ________ times by two plane mirrors placed parallel and facing each other. **[1]**

a) infinite

b) 2

c) 8

d) 4

36. Diffused reflection of light **[1]**

a) May obey or may not obey laws of reflection

b) Occurs from plane surface like mirror

c) obey the laws of reflection

d) does not obey the laws of reflection

37. Human eyes are always paired. What will happen if humans would have only one eye? **[1]**

a) the world would become two-dimensional

b) All the objects will appear faint

for humans

c) Humans will not be able to see objects beyond 90^o

d) All of these

38. A person is not able to view the nearby object clearly but can see far object distinctly. He is suffering from **[1]**

a) Presbyopia

b) Astigmatism

c) Hypermetropia

d) Myopia

39. Images are formed by ________ **[1]**

a) reflection

b) regular reflection

c) diffused reflection

d) irregular reflection

40. Splitting of white light into seven colours is called **[1]**

a) Dispersion

b) Spectrum

c) Rainbow

d) Splitting

41. Which one of the following is the S.I. unit for potential difference? **[1]**

a) Metre

b) Volt

c) Ampere

d) Ohm

42. Rectilinear propagation of light means ________. **[1]**

a) light travels in straight line.

b) light gives us sense of vision.

c) light can bend like curves.

d) light cannot bend like curves.

43. The phenomenon of bending of rays of light when passes from one medium to another medium is called **[1]**

a) Reflection of light

b) Refraction of light

c) Dispersion of light

d) Scattering of light

44. The vision range for a normal human eye is- **[1]**

a) 25 cm to infinity

b) 60 cm from infinite

c) 50 cm from infinite

d) 25 cm from zero

45. The defects of eyes due to absence of cones in retina is **[1]**

a) Astigmatism

b) Poor vision

c) Night blindness

d) Colour blindness

46. Which of the following physical quantity do not require medium for propagation? **[1]**

a) Light

b) Image

c) Speed

d) Sound

47. When light passes through a prism which colour shows maximum deviation? **[1]**

a) Violet colour

b) Red colour

c) Green colour

d) Blue colour

48. Lens in a human eye is ________. **[1]**

a) convex

b) biconvex

c) biconcave

d) concave

49. Rajni is observing her image in a plane mirror. The distance between the mirror and her image is 6m. If she moves 2m towards the mirror, then the distance between Rajni and her image will be ________. **[1]**

a) 8 m

b) 4 m

c) 10 m

d) 5 m

50. For a plane mirror angle between the incident ray and normal is 45^o than the angle of reflection will be: **[1]**

a) More than 45^o

b) Less than 45^o

c) 45^o

d) More or less than 45^o

51. Lens used in spectacles for correction of defects of vision is: **[1]**

a) Always convex

b) Concave or convex

c) Always concave

d) Lens of different shape

52. A student is asked to sit at the last bench because he cannot read the letters written on blackboard clearly from front seat but still is unable to read the letters written in his textbook. Which of the following statements is correct? **[1]**

a) The far point has come closer to him

b) The far point has receded away

c) The near point has receded away

d) The near point has come closer to him

53. Normal is **[1]**

a) Oblique to surface

b) Parallel to surface

c) Perpendicular to surface

d) Convergent to surface

54. A person effected by colour blindners feels problem to identify which two colours? **[1]**

a) Green & Violet

b) Green & Red

c) Black & Blue

d) White and Yellow

55. The path of the light is visible in **[1]**

a) In gaseous medium only

b) In any medium

c) In absence of medium

d) Presence of dispersing medium

56. The angle between incident ray and reflected ray is 120°, then angle of incidence is **[1]**

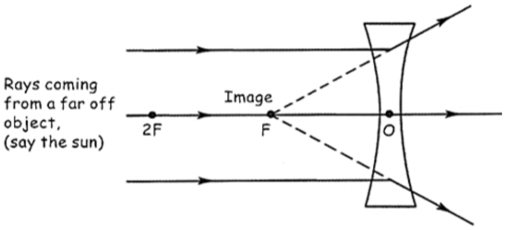
a) 60^o

b) 30^o

c) 90^o

d) 240^o

57. When the object is far off from the lens, the image is ________. **[1]**



a) erect, virtual and enlarged

b) inverted, real and enlarged

c) virtual and smaller

d) erect, virtual and very small

58. The image of an object in case of hypermetropia is formed **[1]**

a) after retina
b) before retina
c) behind the retina
d) at retina

59. A person is standing 2.5m from a plane mirror. How far is the image from the person? **[1]**

a) 2.8m
b) 3m
c) 5m
d) 2.5m

60. In which of the following media, the velocity of light is maximum? **[1]**

a) Water
b) Glass
c) Vacuum
d) Diamond

61. A virtual, erect and magnified image is obtained by a concave mirror when object is placed: **[1]**

a) At a distance greater than twice the focal length.
b) At a distance less than its focal length.
c) At a distance between focal length and twice the focal length.
d) Far off

62. Which of the following has maximum energy? **[1]**

a) Red light
b) Blue light
c) Yellow light
d) Green light

63. A drop of water appears like pearl due to **[1]**

a) total internal reflection of light
b) reflection of light
c) None of these
d) refraction of light

64. Which property of light enables us to see our face in mirror? **[1]**

a) Reflection of light
b) Rectilinear propagation
c) Light is a form of energy
d) Refraction of light

65. Which one of the following statements is INCORRECT based on convex lens? **[1]**

A. Convex lens is thicker in the middle and thinner at the edges.
B. Convex lens is a converging lens.
C. Convex lens is a diverging lens
D. Both Convex lens is thicker in the middle and thinner at the edges and Convex lens is a converging lens

a) Statement C is correct.
b) Statement B is correct.
c) Statement A is correct.
d) Statement D is correct.

66. Which of the following vitamin helps our eye in healthy working? **[1]**

a) Vitamin B
b) Vitamin C
c) Vitamin A
d) Vitamin D

67. Power of a lens is ________. **[1]**

a) $\frac{1}{F}$
b) $\frac{1}{2}$F

c) 2F d) 3F

68. The stars are visible because **[1]**

a) they are away from earth
b) they are absorbing the light of sun
c) they are reflecting the light of sun
d) they are emitting their own light

69. The distance between focus and centre of a curved mirror is known as: **[1]**

a) Focal length
b) Focal point
c) Normal
d) Perpendicular

70. ________ is used in car headlights to reflect the light of the bulb. **[1]**

a) Convex mirror
b) Convex lens
c) Concave mirror
d) Concave lens

71. When two mirrors are placed parallel to each other **[1]**

a) Only two image is formed
b) Large number of image are formed
c) Only image is formed
d) No image is formed

72. If an object is placed between focus and pole of the concave mirror then image formed will be: **[1]**

a) Magnified and erect
b) Inverted and diminished
c) Diminished and erect
d) Magnified and inverted

73. A ray of light is incident on a plane mirror making an angle of 30 degrees with the mirror surface. The angle of reflection for this ray of light will be: **[1]**

a) 45 degrees
b) 0 degree
c) 60 degrees
d) 30 degrees

74. In eye donation, which part of donor's eye is transplanted? **[1]**

a) Cornea
b) Lens
c) Complete lense
d) Retina

75. Primary colours (as per modern colorimetry) consists of **[1]**

a) Red, Green, Blue
b) Red, Yellow, Blue
c) Black, Yellow, White
d) Cyan, Magenta, Yellow